

Ajit Sah

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PROFESSIONAL SUMMARY

- Cybersecurity professional with 3+ years of experience in Security Operations, Incident Response, Threat Detection, Cloud Security, and Enterprise Payment Security. Experienced in alert triage, security monitoring, log analysis, incident investigation, and root cause analysis across high-availability production environments.
- Supported IBM Safer Payments fraud detection platforms and enterprise payment security operations by monitoring system performance, analyzing security events, investigating production issues, and collaborating with cross-functional teams to maintain operational stability.
- Utilized Splunk Enterprise, Grafana, Linux, and AWS security services to support SIEM operations, threat detection, infrastructure monitoring, and incident response activities. Performed security investigations, log correlation, and operational troubleshooting within complex production environments.
- Applied cloud security best practices across AWS services including IAM, EC2, VPC, CloudTrail, and CloudWatch while supporting secure application deployments, security monitoring, and access control management.
- Developed security-focused projects involving AI-assisted SOC operations, prompt injection detection, security automation, and MITRE ATT&CK-aligned investigation workflows using Python and modern security technologies.
- Strong communicator with experience creating operational runbooks, incident response documentation, monitoring procedures, and technical reports that improve investigation efficiency and support security operations.

TECHNICAL SKILLS

SOC & Security Operations: Splunk Enterprise (SPL), Grafana, Alert Triage, Incident Management, Log Analysis, Root Cause Analysis (RCA), Runbook Development, xMatters, High-Availability Monitoring, Production Support.

Fraud & Payment Security: IBM Safer Payments, FastLink Interface (FLI), Transaction Monitoring, Payment Flow Analysis, Clustered Production Environments, Failover Validation.

Cloud Security: AWS IAM, EC2, S3, VPC, CloudTrail, CloudWatch, Security Group, Least-Privilege Access Control, RBAC, Audit Logging.

Vulnerability & Threat Tools: Wireshark, Kali Linux, Nmap, Amass, Metasploit, Vulnerability Scanning, Packet Capture Analysis.

AI/LLM Security AI/ML Security, LLM Security, Prompt Injection & Engineering Defense, AI Security Fundamentals.

Frameworks & Standards: MITRE ATT&CK, NIST CSF, PCI-DSS, OWASP Top 10, Secure SDLC, Defense in Depth, Zero Trust.

Programming & Dev Tools: Python, SQL, Bash, FastAPI, Docker, Git, GitHub, Linux, Vs Code, SecureCRT, MobaXterm, BationServer.

Network & Infrastructure: TCP/IP, DNS, HTTP/HTTPS, Firewall Concepts, VPN Basics, Network Traffic Analysis, OSI Model, Linux CLI.

PROFESSIONAL EXPERIENCE

Production Support Analyst – Payment Security & Fraud Ops

Oct 2023 – Jan 2025

WIT (Contract) Virginia, USA

- Monitor and support enterprise fraud detection and payment processing platforms using IBM Safer Payments in high-volume production environments processing millions of daily transactions, maintaining 99.9%+ platform availability.
- Track p95 latency, transaction throughput, CPU/memory utilization, and TCP connection metrics via Grafana dashboards and Splunk Enterprise proactively identifying degradation patterns before they escalate to incidents.
- Investigate and resolve production security incidents including authorizer timeouts, FastLink buffer overflows, index utilization anomalies, and application disconnects cutting mean time to resolution (MTTR) by ~35%.
- Execute structured incident triage, log analysis, and root cause analysis (RCA) using Linux CLI, Splunk SPL, and application syslogs across clustered high-availability production environments.
- Lead patching cycles, rolling restarts, controlled failover validation, and go-live monitoring for payment security clusters with zero unplanned downtime during critical business windows.
- Authored 10+ troubleshooting runbooks, monitoring SOPs, and incident response procedures reducing on-call escalation rate by 25% and improving cross-team response consistency.
- Coordinate cross-functional incident communication with infrastructure, application, and support teams via xMatters alerting to maintain SLA compliance during major incidents.

Software Developer – Cloud Security

May 2023 – Sep 2024

Braincast Neurotechnology • Hybrid

- Administered AWS environments (EC2, S3, IAM, VPC, CloudTrail, CloudWatch), enforcing least-privilege access control and continuous security monitoring across development and staging environments.
- Configured IAM roles, security groups, and access policies reducing over-privileged accounts by 40% and aligning cloud security posture with CIS AWS Benchmarks.

- ▶ Built Python automation scripts for anomaly detection, log parsing, and suspicious activity alerting via CloudWatch, reducing manual monitoring overhead significantly.
- ▶ Conducted vulnerability assessments using OpenVAS, prioritized findings by severity, and coordinated remediation tracking to support compliance and audit readiness.
- ▶ Implemented API input validation, audit logging, patch management support, and OWASP Top 10 mitigations across all development environments, embedding security into every stage of the SDLC.

Software Developer – Secure SDLC

Oct 2022 – Apr 2023

Maruti Suzuki India Limited • Gurgaon, India

- ▶ Implemented role-based access control (RBAC), input validation, and secure data handling in internal audit and compliance applications serving 500+ users.
- ▶ Participated in CI/CD pipeline testing and application security reviews, identifying access control gaps, injection vulnerabilities, and validation errors during the development of lifecycles.
- ▶ Developed a Vendor Audit Management System using Python, SQL, and HTML with secure SDLC practices, reducing manual audit tracking effort by 30%.
- ▶ Authored technical documentation, onboarding guides, and security workflow procedures to standardize secure development practices across the engineering team.

SECURITY PROJECTS

LLM Prompt Injection Firewall

Mar 2026 - Present

- ▶ Engineered a 3-layer AI threat detection pipeline (regex, semantic embedding similarity, and LLM-as-Judge) to detect and block prompt injection attacks in real time achieving sub-50ms detection latency with zero cloud dependency.
- ▶ Built a SOC-ready audit logging system with SQLite-backed structured logs, threat classification by attack type (instruction override, data exfiltration, role hijacking, delimiter injection), and a live monitoring dashboard replicating core SIEM triage workflows.
- ▶ Researched and implemented defensive controls against OWASP LLM Top 10 vulnerabilities including indirect prompt injection, jailbreak detection, and system prompt extraction demonstrating applied AI threat modeling for blue team operations.

Enterprise AI SecOps Copilot

Sep 2025 – Feb 2026

- ▶ Developed an AI-powered SOC operations platform using Python, FastAPI, and Streamlit integrated with Splunk Enterprise to automate alert triage, incident investigation, and security event summarization.
- ▶ Integrated local LLM models via Ollama to generate automated RCA reports, MITRE ATT&CK technique mapping, and threat investigation summaries reducing analyst triage time on simulated alert sets by ~40%.
- ▶ Engineered a security knowledge base integration layer enabling context-aware AI responses grounded in historical incident data, detection logic, and threat intel simulating a Tier-1 analyst augmentation workflow.

SMBv1 Exploitation & Detection Lab

Mar 2025 – Aug 2026

- ▶ Simulated MS17-010 (EternalBlue) exploit against Windows Server 2008 R2 using Metasploit; captured and analyzed SMBv1 attack traffic in Wireshark to develop automated detection rules improving IDS accuracy by 60% in lab testing.
- ▶ Documented attack telemetry patterns, control gaps, and remediation steps in audit-ready format, supporting threat intelligence use cases and demonstrating technical assessment and reporting skills.

EDUCATION

Washington University of Science and Technology | Alexandria, VA

Jan 2025 – Sept 2026

- ▶ Master of Science Cybersecurity | 4.0 GPA

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- ▶ Microsoft SC-900: Security, Compliance & Identity Fundamentals
- ▶ Hands-on offensive and defensive skills development via TryHackMe active ranked profile
- ▶ CompTIA Security+ Certification (In progress)

Feb 2023 – Dec 2026